

Programming Fundamentals with Python

Python is a high-level, interpreted programming language known for its readability, simplicity, and large ecosystem of libraries. It is widely used in web development, data science, automation, and more.

Variables and Data Types

Variables are used to store data. Python is dynamically typed, meaning you don't need to declare a type explicitly.

```
### Examples of basic data types
name = "Alice"      # string
age = 30            # integer
height = 1.65       # float
is_active = True    # boolean
```

Common data types:

- **int**: Integer numbers
- **float**: Decimal numbers
- **str**: Text (strings)
- **bool**: Boolean (True or False)
- **list**, **tuple**, **dict**, **set**: Collection types

Control Flow

Python uses indentation (usually 4 spaces) to define code blocks.

```
### If-else statement
if age >= 18:
    print("You are an adult.")
else:
    print("You are a minor.")

### Elif (else-if) example
score = 85
if score >= 90:
    print("A")
elif score >= 80:
    print("B")
else:
    print("C")
```